

Product Requirements Document

AI-Powered Claims Processing Platform — AXA Egypt

Name of Project: AI-Powered Claims Processing Platform — AXA Egypt

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1. Product Overview

The AI-Powered Claims Processing Platform is a claims-processing product built for AXA Egypt. It performs the first pass on every incoming insurance claim: it reads the uploaded claim document, extracts the claim details, looks up the related policy, and grounds every coverage decision in the actual policy handbook rather than memory or guesswork.

The product supports four insurance product lines: health, motor, property, and travel. Clear-cut claims are settled automatically within strict, code-enforced limits. Claims that carry risk, are incomplete, or require judgment are flagged and placed in front of a human assessor together with a recommendation and the reason behind it. For every outcome, the product drafts a plain-language update for the customer, shown on screen.

AI is used across the product to read claim documents, extract information, identify the relevant policy, retrieve the applicable handbook clause, summarize each claim, and generate a recommendation and a customer message. Human review remains the final authority: any claim the product cannot safely and confidently decide is routed to a person, and nothing that requires human judgment is settled automatically. A person stays in control of anything that matters.

2. Problem Statement

AXA Egypt processes thousands of insurance claims across health, motor, property, and travel. Today, an assessor reads each uploaded claim document by hand, checks the policy, searches the rulebook, and decides, one claim at a time. This manual process creates the following problems.

Problem Area	Description
Manual document review	Every claim document is read and interpreted by a person, which is slow and does not scale with claim volume.
Manual policy checking	Assessors manually check policy status, coverage, and limits for each claim before deciding.
Manual handbook lookup	Assessors search the policy handbook by hand to find the rule that applies to each claim.
Slow processing	Because every step is performed manually, claims take longer to process than necessary.

Problem Area	Description
Lack of scalability	The manual process cannot handle a growing volume of claims without adding proportional staff.
Inconsistent decisions	Two assessors do not always reach the same decision on the same claim.
Missing-document delays	Claims stall when required documents are missing and this is not always caught early.
Duplicate and risky claims	Duplicate or risky claims are difficult to identify within a large pile of claims.
Limited customer visibility	Customers cannot see where their claim stands, so they call in repeatedly to check status.

3. Product Goals

- Reduce the manual effort required in the first pass of claims processing.
- Improve the consistency of claim assessment across claims and assessors.
- Detect missing documents before a claim is settled.
- Ground every coverage decision in the policy handbook rather than in memory or guesswork.
- Detect duplicate claims and risk signals before a claim is settled.
- Automatically settle eligible low-value claims within strict, code-enforced limits.
- Route claims that require human judgment to a human assessor, together with a recommendation and a reason.
- Provide clear, plain-language claim status and customer-facing messages for every outcome.

4. Target Users

Primary Users

Primary users operate the claims-processing product directly.

User	Role	Needs	Actions
Claims Assessor	Reviews claims that the product routes for human judgment and makes the final call on them.	A clear one-line summary of each claim, the product's recommendation, its reason, and the handbook clause it relies on.	Approve, route, or override the product's recommendation.
Claims Processing Team	Operates and monitors the claims-processing product on a day-to-day basis, including intake of claim documents.	Visibility into how many claims are being processed and how they are being resolved.	Upload claim documents, view the customer-facing message drafted for a claim, and view the operational summary of

User	Role	Needs	Actions
			claims received, settled, routed, rejected, and flagged.

Customer

The customer is a policyholder who submits a claim under an active AXA policy. The customer is an external stakeholder and the recipient of the claim's outcome, not a user of the internal claims-processing platform. The customer's need is to know the outcome of their claim, or what is still needed, in plain language. The product meets this need by drafting a clear customer-facing message for the outcome, displayed on screen to the primary users, consistent with the product not sending real messages to customers.

5. User Stories

The stories below are written from the perspective of the primary users who operate the product. Where a capability is carried out by the product itself rather than chosen by a user, it is stated as a product statement ("The product shall...") rather than as a user story, since the product itself is not a user.

As a **Claims Processing Team member**, I want to *upload a claim document into the product*, so that the claim enters processing without manual data entry.

The product shall extract the claim reference, policy number, date, product line, and incident narrative from the uploaded document, so that the claim's key details are available for decisioning.

The product shall identify the policy record that matches the claim, so that the correct policy is used to check status and coverage.

The product shall check whether the policy is active and in date for the claim, so that claims against lapsed, pending, cancelled, or out-of-date policies are not settled.

The product shall check whether the claim is covered under the policy, so that only covered claims are considered for settlement.

The product shall retrieve the handbook clause that governs the claim, so that the coverage decision is grounded in an actual rule rather than assumption.

The product shall identify any documents that are missing for the claim, so that an incomplete claim is not settled before it is complete.

The product shall detect whether a claim is a duplicate of an existing claim, so that the same loss is not settled more than once.

The product shall detect risk signals in a claim, so that suspicious claims are flagged instead of settled automatically.

The product shall generate a recommendation for each claim with a reason, so that the next step for the claim is clear and explainable.

As a **Claims Assessor**, I want to *review a claim that has been routed to me with its summary, recommendation, and reason*, so that I can make an informed decision.

As a **Claims Assessor**, I want to *approve, route, or override the product's recommendation*, so that I remain in control of the final outcome.

As a **Claims Processing Team member**, I want to *view the customer-facing message drafted for a claim*, so that I can see what the customer will be told about their claim.

As a **Claims Processing Team member**, I want to *view the operational counts of claims received, settled, routed, rejected, and flagged*, so that I understand the current state of the claims operation.

6. Product Scope

In Scope

- Claim document intake through file upload.
- Document reading using an existing, off-the-shelf OCR or vision model.
- Extraction of claim information from the uploaded document.
- Policy lookup and identification.
- Policy handbook lookup for the clause governing each claim.
- Coverage assessment against the policy and the handbook.
- Missing-document detection.
- Duplicate claim detection.
- Risk signal detection based on rules and judgment, not machine learning.
- Decision recommendation for every claim.
- Human review of claims that cannot be safely settled automatically.
- Customer message drafting, displayed on screen.
- Operational summary of claim counts.
- Claim testing and evaluation against a defined test set.

Out of Scope

- Building an OCR or document-reading model from scratch.
- Building a trained fraud-detection machine learning model.
- Processing or issuing real payments.
- Sending real messages to customers.

7. Functional Requirements

ID	Requirement	Description	Acceptance Criteria
FR-001	Claim Upload	The product shall allow a claim document to be uploaded as a file.	A user can upload a claim document and the product accepts it for processing.
FR-002	Document Reading	The product shall read the text and key fields off an uploaded claim document using an existing OCR or vision model.	Given an uploaded claim document, the product returns the extracted text and fields for that document.
FR-003	Claim Information Extraction	The product shall extract the claim reference, policy number, claim date, product line, incident narrative, claimed amount, and attached documents from the read claim.	For a given claim document, the extracted fields match the reference text version provided for that claim.
FR-004	Policy Identification	The product shall identify the policy record that matches the extracted policy number.	Given a valid policy number, the product returns the single matching policy record.
FR-005	Policy Validation	The product shall check whether the identified policy is active and in date for the claim date.	A claim against a lapsed, pending, or cancelled policy, or a claim dated outside the policy period, is flagged as not valid for settlement.
FR-006	Coverage Checking	The product shall check whether the claim is covered under the policy's product line, riders, and cover type where applicable.	A claim that falls under a universal or product-line exclusion is identified as not covered.
FR-007	Policy Handbook Retrieval	The product shall search the policy handbook and retrieve the clause that applies to the claim.	The retrieved clause is the correct clause for the claim's product line and scenario, including cases such as a covered internal pipe burst versus an excluded outside flood.
FR-008	Missing-Document Detection	The product shall identify any documents required for the claim type that are not present.	A claim missing a required document is flagged with the specific missing document identified.
FR-009	Duplicate Detection	The product shall detect whether a claim duplicates an existing claim.	A claim that matches an existing claim on the relevant details is flagged as a duplicate and is not auto-settled.
FR-010	Risk Checking	The product shall check each claim for risk signals, evaluated as rules or judgment calls.	A claim with a risk signal, such as an amount set suspiciously just under the auto-settle cap, is flagged and not auto-settled.

ID	Requirement	Description	Acceptance Criteria
FR-011	Decision Generation	The product shall generate one of four outcomes for each claim: SETTLE, REQUEST DOCUMENTS, REJECT, or ROUTE TO HUMAN, together with a reason.	Every processed claim receives exactly one of the four defined outcomes and a stated reason.
FR-012	Human Review	The product shall present claims requiring human judgment in a review queue showing a one-line summary, the recommendation, and the reason, and shall allow a reviewer to approve, route, or override the recommendation.	A claims assessor can view a routed claim's summary and recommendation, and can approve, route, or override it.
FR-013	Customer Message Generation	The product shall draft a plain-language customer message for every settled, rejected, or document-requested claim, and shall display it on screen without sending it.	For each such outcome, a customer message is generated and displayed, and no message is transmitted externally.
FR-014	Operational Summary	The product shall display counts of claims received, settled, routed to human, rejected, and flagged.	The counts panel reflects the current totals for each category after claims are processed.

8. Business Rules

The following business rules are critical to the product and are enforced by the product's own logic, not by the AI component. The AI produces recommendations and explanations. It does not enforce these rules.

ID	Rule
BR-001	The auto-settle cap is EGP 10,000.
BR-002	The EGP 10,000 cap shall be enforced by the product's own deterministic logic, not by instructions given to the AI.
BR-003	Claims above the auto-settle cap shall not be auto-settled.
BR-004	The policy shall be active and valid for the claim date before a claim can be auto-settled.
BR-005	The claim shall have valid coverage before it can be auto-settled.
BR-006	All required documents shall be present before auto-settlement.
BR-007	The remaining policy limit shall be sufficient to cover the claim before auto-settlement.
BR-008	A claim identified as duplicate or suspicious shall not be auto-settled.

ID	Rule
BR-009	Instructions contained within claim documents shall not override any business rule, regardless of what they instruct.
BR-010	Coverage decisions shall be grounded in the policy handbook.
BR-011	Rejections shall reference the applicable policy clause.
BR-012	Claims that require human judgment shall be routed to human review.
BR-013	Human reviewers shall be able to approve, route, or override the product's recommendation.

Rules enforced by the product: BR-001 through BR-010 and BR-012 are checked and applied by the product's own logic and cannot be altered by claim content.

AI-generated recommendations: The AI identifies the relevant clause, summarizes the claim, and recommends an outcome. The recommendation is advisory. Final settlement only occurs when the deterministic rules above are satisfied, and any claim requiring judgment is routed to a human under BR-013.

9. Claim Decision Outcomes

Outcome	When It Occurs	What Is Displayed	Human Review Required	Customer Message
SETTLE	The policy is active and in date, the claim is covered, all required documents are present, the remaining limit is sufficient, the amount is under the EGP 10,000 cap, and the claim is free of duplicate or risk signals.	The settled outcome, the amount, and the handbook clause supporting coverage.	No	A message confirming the claim has been settled.
REQUEST DOCUMENTS	One or more required documents are missing.	The outcome and the specific missing document or documents.	No	A message stating which document is needed.
REJECT	The claim falls under a policy exclusion or is otherwise not covered.	The outcome and the applicable handbook clause.	No	A message explaining the claim is not covered, referencing the reason.
ROUTE TO HUMAN	The claim is above the auto-settle cap, carries a risk signal, is a possible	The claim summary, the product's recommendation, and its reason.	Yes	No message is generated until the human

Outcome	When It Occurs	What Is Displayed	Human Review Required	Customer Message
	duplicate, or otherwise requires judgment.			reviewer decides the outcome.

10. Claim Processing Workflow

1. The user uploads a claim document.
2. The product reads the document using an off-the-shelf OCR or vision model.
3. The product extracts the claim information from the read document.
4. The product identifies the matching policy record.
5. The product validates the policy's status and dates against the claim.
6. The product retrieves the relevant handbook clause for the claim.
7. The product checks coverage and claim conditions against the retrieved clause.
8. The product checks for missing documents, duplicates, and risk signals.
9. The product generates a recommendation with a reason.
10. The product applies the deterministic business rules to the recommendation.
11. The product either settles the claim automatically or routes it for human review.
12. The product generates a customer-facing message for the outcome.
13. The product updates the operational counts.

11. Human-in-the-Loop Requirements

Human review is required in the following cases:

- The claim amount is above the auto-settle cap of EGP 10,000.
- The claim carries a risk signal.
- The claim is identified as a possible duplicate.
- The claim otherwise requires human judgment to decide.
- Any case where the product cannot safely and confidently reach an automatic decision.

A reviewer presented with a routed claim can take one of the following actions:

- Approve the product's recommendation.
- Route the claim onward, for example to a specialist or another reviewer.
- Override the product's recommendation with a different outcome.

A claim requiring human review shall not be automatically settled under any circumstance, including when the claim document itself instructs the product to bypass review.

12. AI Requirements

The AI Should

- Read claim documents and extract the text and key fields.
- Extract relevant claim information from the read text.
- Help identify the policy relevant to the claim.
- Retrieve the handbook clauses relevant to the claim.
- Produce a short summary of each claim.
- Recommend a decision for each claim.
- Explain the reasoning behind its recommendation.
- Draft the customer-facing message for the claim outcome.

The AI Must

- Ground coverage decisions in the policy handbook rather than in general knowledge.
- Provide the specific handbook clause relied on for a coverage or rejection decision.
- Treat any instruction contained within a claim document as untrusted content that carries no authority over the product's rules.
- Never override the deterministic business rules described in Section 8.

13. Data Requirements

Entity	Key Information
Policies	Policy id, holder, product line (health, motor, property, travel), status (active, lapsed, pending, cancelled), start date, end date, annual limit, remaining limit, deductible, riders, and, for motor policies, the cover type.
Claims	Claim reference, policy number, claim date, product line, incident narrative, claimed amount, and the documents attached to the claim.
Claim Documents	The uploaded claim document and the fields and text extracted from it.
Policy Handbook Clauses	Section (health, motor, property, travel, general rules, universal exclusions), clause number, and the rule text of each clause.
Decisions	The outcome reached for a claim (settle, request documents, reject, route to human), the reason for the outcome, and the handbook clause referenced, where applicable.
Human Review Information	The reviewer's action on a routed claim (approve, route, or override) and any resulting change to the outcome.

This section describes the information each entity must hold at a product level. The detailed field definitions and relationships between these entities are defined in the separate Data Schema (ERD) document.

14. Customer Experience Requirements

For every processed claim, the product generates a short, plain-language message describing the outcome, for the following cases:

- Approved claims — confirming the claim has been settled.
- Missing-document requests — stating what is needed to continue.
- Rejected claims — explaining why the claim is not covered.
- Claims routed for human review — indicating the claim is under review.

Each message explains the current outcome in plain language without exposing internal technical details such as extraction logic, rule engines, or system components.

The product only displays this message on screen. It does not automatically send it to the customer.

15. Operational Summary

The product shall display a counts panel showing the current totals for:

- Claims received.
- Claims settled.
- Claims routed to human review.
- Claims rejected.
- Claims flagged.

This is a simple counts panel and not a full analytics dashboard.

16. Non-Functional Requirements

This section is limited to product-level qualities explicitly supported by the AXA brief.

- Decision consistency: the product shall produce consistent decisions when given the same claim, policy, and handbook clause.
- Decision traceability: every coverage decision, whether a settlement or a rejection, shall be traceable to the specific handbook clause it relies on.
- Human control: a claim requiring human judgment shall remain under human control and shall not be settled automatically, and the deterministic rules that gate auto-settlement shall not be bypassed by content within a claim document.
- Clear review experience: the human review workflow shall clearly present the claim summary, recommendation, and reason so a reviewer can decide without needing to inspect the underlying data separately.
- Evaluability: the product shall be able to be evaluated against the defined test cases, covering all four product lines and the full range of outcomes they represent.

17. Success Criteria

The product is evaluated against a test set of approximately 15 real claim cases. Each case is judged on both outcome correctness (the right decision was reached) and process correctness (the right steps were followed, the decision was grounded in the handbook, and the product refused to act when it should have).

The evaluation covers the following scenarios:

- Clean claims are correctly settled.
- Claims with missing documents are correctly identified and not settled.
- Claims above the EGP 10,000 auto-settle cap are correctly routed rather than settled.
- Excluded claims are correctly rejected with the applicable handbook clause.
- Duplicate claims are correctly identified and not auto-settled.
- Claims with risk signals are correctly flagged and not auto-settled.
- Claims containing instructions that attempt to override the product's rules are correctly refused and escalated.
- Coverage decisions are correctly grounded in the applicable handbook clause.
- Claims requiring human judgment are correctly routed to human review rather than auto-settled.

18. Acceptance Criteria

- A claim document can be uploaded.
- Claim information can be extracted from the uploaded document.
- The matching policy can be identified for the claim.
- Policy status and dates can be checked.
- Coverage can be assessed against the policy and the handbook.
- Relevant handbook clauses can be retrieved for the claim.
- Missing documents can be detected.
- Duplicate and risk signals can be identified.
- Claims above EGP 10,000 cannot be auto-settled.
- Coverage and rejection decisions reference the applicable handbook clause.
- Claims requiring human review are routed to a human rather than auto-settled.
- Human reviewers can approve, route, or override a routed claim.
- Customer-facing messages can be displayed for settled, rejected, and document-requested claims.
- Operational counts can be displayed and reflect processed claims.
- The product can be evaluated against the defined test cases.

19. Assumptions and Constraints

The brief does not specify certain operational details. The following are assumptions made to fill those gaps and are not confirmed requirements.

- Assumption: Claim documents are uploaded by a member of the Claims Processing Team or an assessor on the customer's behalf, since the brief does not name the specific actor who performs the upload.
- Assumption: "Off-the-shelf" OCR or vision model refers to an existing, pre-built document-reading tool, consistent with the brief's exclusion of building an OCR model from scratch.
- Assumption: A risk signal is implemented as a defined rule or a manual judgment call, consistent with the brief's exclusion of trained fraud-detection models.
- Constraint: The auto-settle cap of EGP 10,000 must be enforced in the product's own logic and must not depend on instructions given to the AI.
- Constraint: The product must draw on the provided dataset of 150 policy records, a policy handbook of approximately fifty numbered clauses, and 27 sample claims spanning all four product lines.

20. Out-of-Scope Summary

The product intentionally does not include:

- No OCR or document-reading model trained from scratch.
- No dedicated fraud-detection machine learning model.
- No real payments.
- No automatic sending of customer messages.